

Assignment 1: Overview of hardware component theory.

Task 1: Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

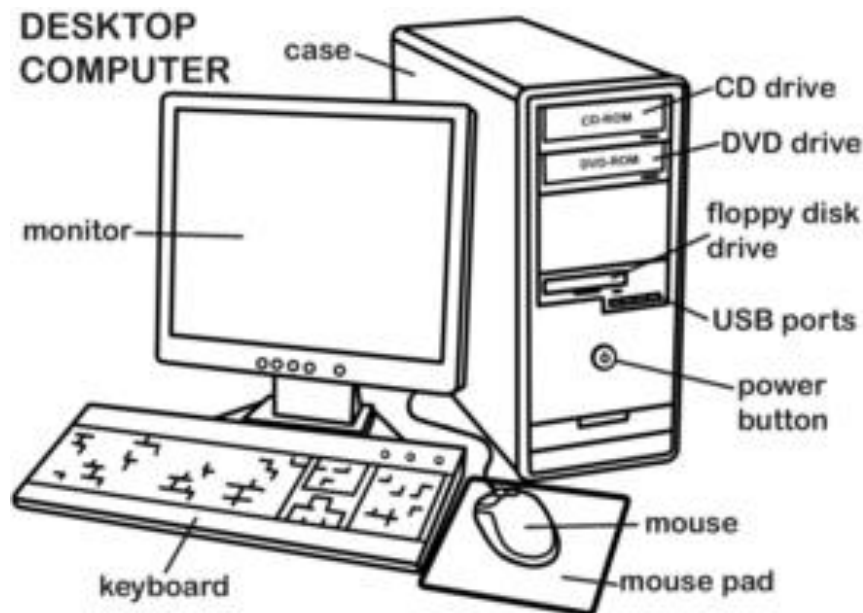


Fig. 1. Desktop PC

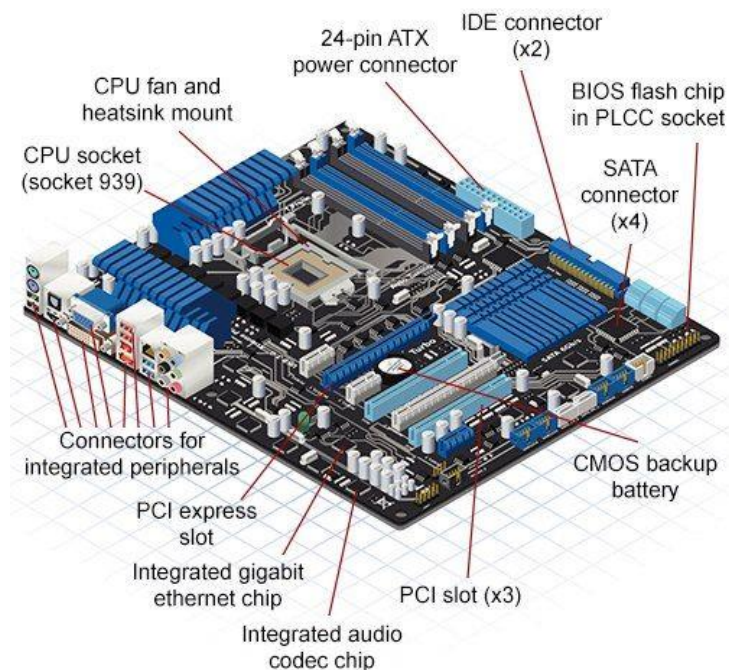


Fig. 2. Motherboard



Fig. 3. CPU

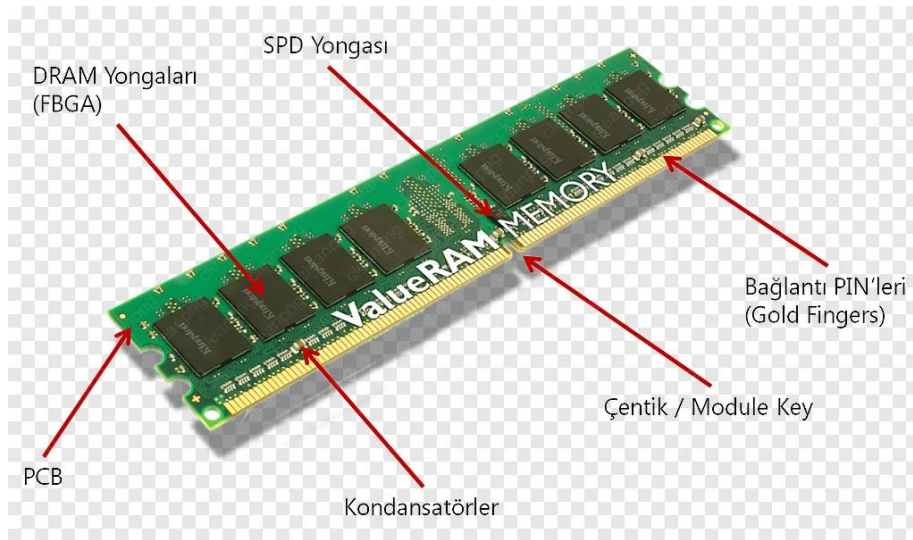


Fig. 4. RAM

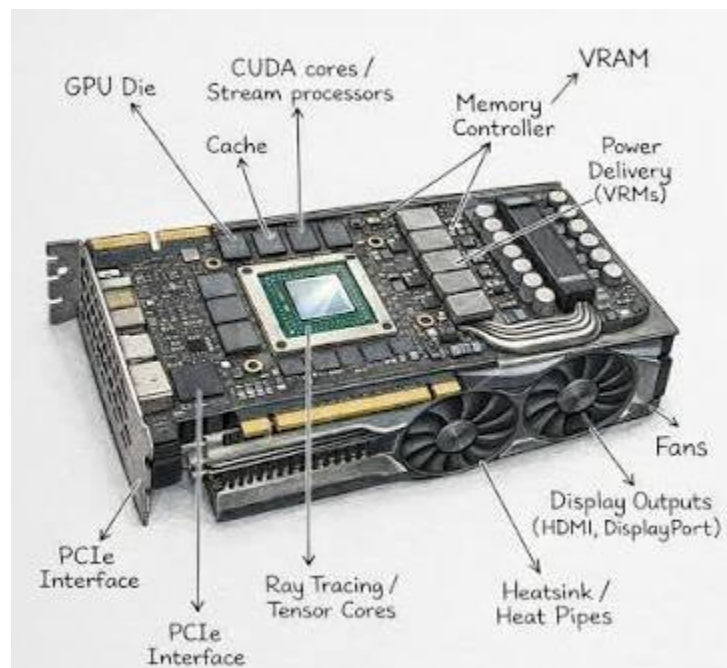


Fig. 5. GPU

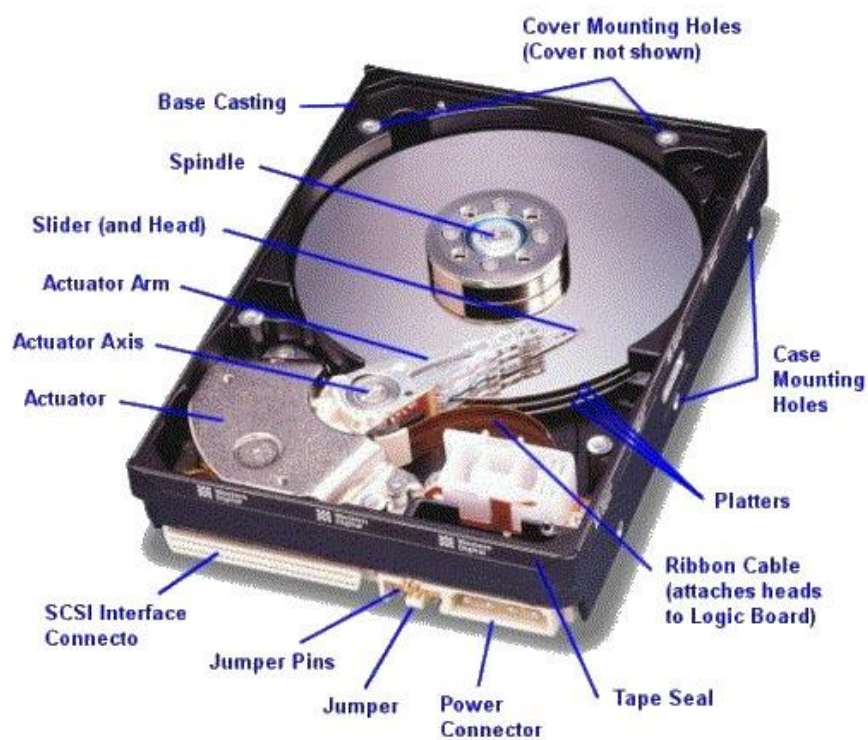


Fig. 6. HDD

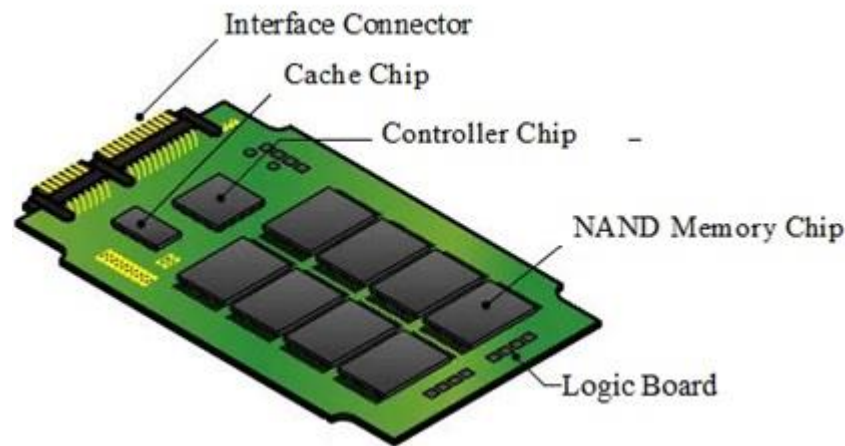


Fig. 7. SSD

Task 2: Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases. Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

HDD	SSD
Slower read/write speed	Much faster read/write speed
Less durable due to moving parts	More durable because it has no moving parts
Easily damaged by shock or drops	Highly shock-resistant
Produces noise while operating	Silent operation
Uses more power	Uses less power
Larger capacity at a lower cost	Usually, smaller capacity at a higher cost
Desktop PCs, budget computers, mass data storage, backups	Laptops, gaming PCs, high-performance desktops, fast boot drives
Cheaper per GB	More expensive per GB

Task 3: Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU: The CPU is the brain of the computer. It processes game instructions, controls player actions, and manages communication between different hardware components. A faster CPU helps games like PUBG or Free Fire run more smoothly.

RAM: RAM temporarily stores the game data that the CPU needs while the game is running. More RAM allows the game to load faster, reduces lag, and helps you switch between the game and other programs smoothly.

GPU: The GPU is responsible for rendering the game's graphics, including characters, maps, lighting, and visual effects. A powerful GPU provides higher frame rates, smoother gameplay, and better image quality in games like PUBG or Free Fire.

Task 4: Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

CPU (Processor): Upgrade to a faster and newer CPU so the game can process player movements, AI, physics, and match calculations more efficiently. This improves overall performance and reduces stuttering.

GPU (Graphic Card): Upgrade to a more powerful graphics card to render high-quality graphics, realistic stadiums, player animations, and visual effects. A better GPU increases frame rates (FPS) and allows you to play at higher graphics settings.

RAM: Increase the RAM to at least **16 GB** (or more if needed). More RAM allows the game to load data quickly, reduces lag, and enables smooth multitasking while gaming.

SSD (Solid State Drive): Replace or add an SSD to install the game. An SSD significantly reduces boot time, game loading screens, and update installation time compared to an HDD.

Power Supply Unit (PSU): If you install a more powerful CPU or GPU, you may also need a higher-wattage, reliable PSU to provide stable power and ensure the system runs safely.

Cooling System (CPU Cooler/ Case Fans): Upgrading the cooling system helps keep the CPU and GPU at safe temperatures during long gaming sessions. Better cooling prevents overheating and maintains consistent performance.

Motherboard: If the new CPU or RAM is not compatible with the current motherboard, you may need to upgrade the motherboard to support the latest hardware and technologies.